CRYOMECH

Cryocooler

PT420RM - CPA1114

INSTALLATION and OPERATION MANUAL

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Section 1

Overview

Section 1: Overview

This section provides a general overview of a cryocooler, the organizational structure of this manual, and a glossary of defined terms and acronyms.

1.1 Cryocooler system

1.1.1 General description

The operation of a Cryomech cryocooler is based on a closed-loop helium expansion cycle. The cryocooler consists of a helium compressor package, cold head, flex lines and cold head motor cord.

The compressor package receives low pressure helium from the cold head, pressurizes the helium gas and returns high pressure helium gas to the cold head via flex lines. The cold head reaches cryogenic temperatures through adiabatic expansion by using the high pressure helium gas from the helium compressor package and discharging low pressure helium gas and returning it to the compressor package.

1.1.2 Organization of the manual

This manual is divided into eight section that provide the information needed to install, operate and perform maintenance on a Cryomech cryocooler. It is important to note that bulleted list may be performed in any order but numbered lists (labeled 1, 2, etc.) are representative of sequential actions that must be performed in the order listed.

1.2 Glossary

1.2.1 Definitions

Aeroquip Couplings:

The term "Aeroquip couplings" is used generically to describe the self-sealing fittings that connect components, e.g. helium flex lines are connected to the helium compressor package and to the cold head via Aeroquip couplings.

Category II Installation:

Category II refers to the potential for transient over-voltage conditions in the mains power connection to the equipment. See IEC 664, Sub-clause 5.6 for further details.

Closed Loop System:

This refers to a cryogenic system that has no helium loss.

Cold Head:

The cold head is an expansion device which is capable of reaching cryogenic temperatures.

Cold Head Motor Cord:

The cold head motor cord provides electrical power from the helium compressor package to the cold head motor.

Cold Head Heat Exchanger:

The heat exchanger(s) on the cold head provide cooling at cryogenic temperatures by transferring heat to the helium within the system.

Cryogenic Temperatures:

Temperatures lower than 120 K or -153 °C.

Cryocooler:

A cryocooler is a cryogenic refrigeration system based upon a closed loop helium expansion cycle. It consists of a helium compressor package, helium flex lines, and a cold head (expansion device).

Gifford-McMahon (GM):

A cold head which expands the helium using a displacer.

Helium Compressor Package:

Provides system safety controls, houses the components that compress, cool and purify the helium gas and provides electrical interface controls.

Helium Flex Lines:

The helium flex lines are corrugated, stainless steel hoses that transport helium between the helium compressor package and the cold head.

Pollution Degree 2:

Pollution degree 2 refers to the extent to which the local environmental conditions could affect the electrical safety of the system. See EN 61010 or UL 61010A for further details.

Pulse Tube:

A cold head that does not use a displacer or piston.

System:

The term "system" is used as a synonym for cryocooler. It consists of a compressor package, helium flex lines and cold head.

1.2.2 Acronyms

The following acronyms are used in the text and provided here for convenient lookup:

FPT Female Pipe Thread

GPM Gallons per Minute

LPM Liters per Minute

MPT Male Pipe Thread

psig Pounds per Square Inch Gauge

Section 2

Warranty

Section 2: Warranty

2.1 Statement of warranty

Provided the customer installs, operates and maintains this cryocooler in accordance with the specifications and procedures set forth in this manual, Cryomech, Inc. extends a limited warranty on parts and workmanship. This warranty covers all non-user serviceable components of the compressor package, cold head and helium flex lines. The warranty does not cover user-serviceable electronics (such as fuses).

If found to be defective and in accordance with the terms of the limited warranty, Cryomech, Inc. will provide warranty replacement parts at no cost to the customer. Cryomech, Inc. does not cover onsite service. Defective components must be returned to Cryomech, Inc. for service. Customers are responsible for all shipping and handling charges associated with each warranty repair.

GM cryocoolers have a warranty period of three (3) years or 8,000 operating hours, whichever comes first. The GM cold head is warranted for the same three (3) years or 8,000 operating hours to deliver the specified temperatures and cooling capacities.

Pulse tube cryocoolers have a warranty period of three (3) years or 12,000 operating hours, whichever comes first. The pulse tube cold head is warranted for the same three (3) years or 12,000 operating hours to deliver the specified temperatures and cooling capacities.

2.2 Conditions that can void the warranty

- Operation of this cryocooler in situations that do not meet the specifications in this manual will void the warranty. If you intend to operate the cryocooler outside any of the specified conditions, prior written authorization from Cryomech, Inc. must be granted.
- Installing any software on a Cryomech, Inc. product that was not provided by Cryomech, Inc. will void the warranty. See Section 5
- See Section 6 for installation guidelines. Failure to follow these guidelines will void the warranty.
- Exceptional care must be used when handling the cold head. The warranty will not cover dented or bent tubes.
- Heating any part of the cold head above 325 K (125 ºF, 52 ºC) will cause damage and void the warranty.
- With the cold head oriented vertically, the maximum allowable load cannot exceed the value specified in Section 5. Excessive loading will damage the cold head. This type of damage is not covered by the warranty. If your application requires a load heavier than the weight specified in Section 5, prior written authorization from Cryomech, Inc. must be granted.
- Mounting the cold head in an orientation other than vertical will void the warranty without prior written authorization from Cryomech, Inc.
- Alterations to the cryocooler are prohibited without prior written authorization from Cryomech, Inc. Modifications without prior written approval will void the warranty.
- Cooling water must meet the specifications in Section 5. If non-conforming cooling water is introduced into the system, even for cleaning purposes, the warranty will be void.

- The mains supply voltage to the cryocooler must be within 10% of the nominal voltage specification in Section 5. Operating the cryocooler outside the voltage specification will void the warranty.
- Do not allow the flex lines to contact any corrosives or commercial cleaning agents. Flex line damage caused by exposure to corrosives or commercial cleaning agents will not be covered by the warranty.
- Applying extreme bend stress to a flex line will cause permanent damage. Bending a flex line to less than a 12 in (30 cm) radius will void the warranty. See Section 6.
- Operating the cold head in a magnetic field of greater than 500 gauss will void the warranty.
- Attempting to disassemble the cold head will void the warranty. Contact Cryomech, Inc. if the cold head needs to be serviced. See Section 8.
- When adding helium to the cryocooler, the helium must be at least 99.999% pure. Adding helium that is less than 99.999% pure will contaminate the cryocooler and void the warranty. See Section 8.
- Consumable, non-electrical, items such as adsorbers must be replaced with parts provided by Cryomech, Inc. Replacing consumable items with third party produced products will void the warranty for all parts of the cryocooler, including the helium compressor package, flex lines and cold head. See Section 8.
- Excluding the designated water cooling circuit, never allow water to contact any of the components of the system. Exposing the helium compressor package, flex lines or cold head to water will void the warranty.

Section 3

Safety

Section 3: Safety

3.1 Safety and information symbols

3.1.1 Equipment symbols

The safety and information symbol stickers placed on Cryomech, Inc. cryocoolers are defined below.

	Alternating current. The symbol signifies that alternating current is present.
	Internal ground. This symbol represents an internal protective grounding terminal. Such a terminal must be connected to earth ground prior to making any other connections to the equipment.
	Warning Icon. Refer to the documents that accompany the equipment.
	The CE label is placed on a product if the product has been tested for and meets the safety standards set by the European Community. CE stands for Conformité Européenne. All pressure equipment bearing the CE label has a maximum temperature of 94 °C (200 °F) and a maximum pressure 29 barg (425 psig).
	The TUV label is placed on a product if the product has been tested for and meets the safety standards set by the US and Canada. The cTUVus mark is officially recognized as an equivalent and direct replacement of the UL and CSA marks. Nation Recognized Testing Laboratories (NRTL) like TÜV Rheinland can test to the same standard and specifications as those defined by UL and CSA.
	Power switch. This symbol designates an in/out or push/push switch.
	Read the manual or handbook sign. When this symbol is found on a piece of equipment, the user should read the whole manual before starting installation or use. This symbol is found on the compressor package.

3.1.2 Icons in the manual

Definitions of Warning and Caution and Information icons in the manual

	Warning Icon. A warning message is used when failure to observe instructions or precautions could result in injury or death to humans.
	Electrical Warning Icon. An electrical warning message is used when failure to observe instructions or precautions could result in electrical shock or burns to humans.
	Caution Icon. A caution message is used when failure to observe instructions or precautions could result in significant damage to equipment and/or facilities.
	Information Icon. The accompanying message contains information to aid the operator in obtaining the best performance from the equipment or other important information that does not involve danger to equipment or humans.

3.2 Warnings and cautions

The warnings and cautions for the Cryomech cryocooler are listed here by section. The same warnings and cautions appear in each respective section.

3.2.1 Section 5: Specifications

Section 5.2 Technical specifications

CAUTION

Operation of the cryocooler in any situation that does not meet the specifications in this section will void the warranty. If you plan to operate the system outside any of the specified conditions, contact Cryomech, Inc.

CAUTION

The use of DI (De-Ionized) water is prohibited. DI water causes damage to the compressor package and will not be covered under warranty.

3.2.2 Section 6: Installation

Section 6.1 Cold head installation

CAUTION

Failure to follow these installation guidelines will void the warranty.

CAUTION

Special care must be taken when handling the cold head. The warranty will not cover dented or bent tubes.

CAUTION

Operating the cold head in a magnetic field of greater than 500 gauss will void the warranty.

Section 6.1.2 Initial preparation of the cold head

CAUTION

Heating any part of the cold head above 325 K (125 ºF, 52 ºC) will cause damage and void the warranty.

CAUTION

With the cold head oriented vertically, the maximum allowable load cannot exceed the value specified in Section 5. Excessive loading will damage the cold head. This type of damage is not covered by the warranty. If your application requires a load heavier than the weight specified in Section 5, prior written authorization from Cryomech, Inc. must be granted.

CAUTION

Mounting the cold head in an orientation other than vertical will void the warranty without prior written authorization from Cryomech, Inc.

CAUTION

Alterations to the cryocooler are prohibited without prior written authorization from Cryomech, Inc. Modifications without prior written approval will void the warranty.

Section 6.1.6 Connect the remote motor assembly's flex line- Remote Motor Cold Heads Only

CAUTION

Follow this procedure carefully when connecting and disconnecting the helium flex lines. Failure to follow the procedure will void the warranty.

CAUTION

Do not allow the flex lines to contact any corrosives or commercial cleaning agents. Flex line damage caused by exposure to corrosives or commercial cleaning agents will not be covered by the warranty.

CAUTION

Applying extreme bend stress to a flex line will cause permanent damage. Bending a flex line to less than a 12 in (30 cm) radius will void the warranty.

Section 6.2 Compressor installation

CAUTION

Failure to follow these installation guidelines could result in voiding the warranty.

Section 6.2.1 Prepare the compressor package location

WARNING

The helium compressor package must be positioned to provide easy access to the front panel mounted circuit breaker.

Section 6.2.2 Connect the water lines to the compressor

CAUTION

Cooling water must meet the requirements in Section 5. Introduction of cooling water that does not meet the specifications in Section 5, even for cleaning purposes, it will void the warranty.

CAUTION

Do not apply heat to the cooling water inlet and outlet connectors located on the front panel of the helium compressor package.

Section 6.2.3 Connect the compressor package to the main power

CAUTION
The mains supply voltage to the helium compressor package must be within 10% of the nominal voltage specification in Section 5. Operating the helium compressor package outside the voltage specification will void the warranty.

WARNING
Follow all local electrical codes and guidelines when during installation.

WARNING
One lead of the helium compressor package is grounded. Never bypass this ground or attach the helium compressor package to an ungrounded circuit. A dangerous electrical hazard will develop.

Section 6.2.4 Connect the helium flex lines to the cold head or remote motor assembly and the compressor package

CAUTION
Follow this procedure carefully when connecting and disconnecting the helium flex lines. Failure to follow the procedure will void the warranty.

CAUTION
Do not allow the flex lines to contact any corrosives or commercial cleaning agents. Flex line damage caused by exposure to corrosives or commercial cleaning agents will not be covered by the warranty.

CAUTION
Applying extreme bend stress to a flex line will cause permanent damage. Bending a flex line to less than a 12 in (30 cm) radius will void the warranty.

3.2.3 Section 7: Operation

Section 7.1.1 Checks before operating

CAUTION
A voltage deviation of more than 10% above or below the voltage rating can cause helium compressor motor overheating and possible failure. Indications of voltage operation outside that range will void the helium compressor warranty.

CAUTION
Operating the cold head in a magnetic field of greater than 500 gauss will void the warranty.

Section 7.1.2 Startup procedure

WARNING
The system operates in a closed loop cycle at pressures higher than 200 psig (12.8 bar) and at temperatures below 120 K.

Section 7.6 Disassembling the system for transport or storage

WARNING
Do not touch the cold end of the cold head until it has warmed to room temperature. If there is frost on the cold head, it is too cold to touch.

3.2.4 Section 8: Routine maintenance

Section 8.3 Cold head

CAUTION
The Pulse Tube style cold heads contain no user-serviceable parts. Attempting to disassemble the cold head will void the warranty. Contact Cryomech, Inc if the cold head needs to be returned for servicing.

CAUTION
Service to Gifford-McMahon style cold heads must be performed using Cryomech, Inc parts and the appropriate work instructions. Contact Cryomech, Inc. before servicing the cold head.

Section 8.4 Replace the adsorber

CAUTION
At no time should the Aeroquip couplings be removed from the adsorber. Replacement can be completed without relieving system pressure, since the adsorber is equipped with Aeroquip couplings for sealed removal.

WARNING
The input power must be disconnected from the helium compressor package before removing any helium compressor package panels.

Section 8.5 Vent excess helium from the helium compressor package

CAUTION
Venting more than 5 psig (0.34 bar) of helium per minute will lead to improper oil migration within the system. If this condition occurs, factory service will be required.

Section 8.6 Recharge helium to the helium compressor package

CAUTION

When adding helium to the helium compressor package, the helium must be 99.999% pure. Contamination of the helium by other gases is a common cause of premature failure and is not covered by the warranty.

CAUTION

No more than 5 psig (0.34 bar) of gas should be added per minute to prevent improper oil migration within the system. If such contamination occurs, factory service will be required.

Section 8.7 Cleaning

CAUTION

Excluding the designated water cooling circuit, never allow water to contact any of the components of the system. Exposing the system to water will void the warranty.

WARNING

Never remove an Aeroquip coupling from the helium flex lines, cold head, adsorber, or helium compressor package without first relieving the helium charge. Pressure in any of the components can cause the coupling to become a projectile that may cause injury.

Section 4

Inspection and Unpacking

Section 4: Inspection and Unpacking

IMPORTANT

If there is any visible damage to the packing crates or any components contact Cryomech, Inc. immediately.

4.1 Inspection of crate

Be sure to note on the shipping documents any visible damage to the crate, including tip indicators that have been activated.

4.2 Unpacking

The system is shipped in a secure packing crate. The base of the packing crate is a wooden pallet, to which the system is strapped. The walls of the crate are placed around the system, attached to the pallet, and to each other with tension clips (Klimp fasteners) and screws. After adding packing material, the top is clipped onto the packing crate.

4.2.1 Directions for unpacking:

1. After ensuring there is no visible damage to the packaging, remove the top of the packing crate by unfastening the Klimp fasteners that join the top to the sides.
2. Check for tip indicators on the inside and outside of the packing crate.
3. Check for any visible signs of damage besides activated tip indicators.
4. Locate and remove items that can easily be lifted out of the crate.
5. Remove the sides of the packing crate by unfastening the Klimp fasteners and screws that join the sides to the pallet.
6. Remove packing material and any straps that anchor items to the pallet. Retain the packing materials and crate for use in the future, to ship the equipment to Cryomech, Inc.
7. Ensure that adequate space is prepared for the equipment (see Section 6 for installation directions) before continuing.

4.3 Inspection of equipment

4.3.1 Packing list

There is a packing list included with the system. Check that all parts listed on the packing list are included in the crates. Then, check for any signs of damage to the parts.

4.3.2 Cold head

1. See weights and dimensions of the cold head in Section 5 to ensure components can be moved safely.
2. Inspect the box that contains the cold head for any signs of physical damage.
3. Remove the screws holding the cover to the box and lift the cover.
4. Carefully remove packing material to expose the cold head. Retain the packing material for use in the future, to ship the equipment to Cryomech, Inc.
5. Use all equipped eyebolts for lifting the cold head. Follow local regulations while lifting or hoisting the cold head.
6. Carefully remove the cold head from the box and place on a clean, secure work surface.

7. Inspect the cold head for any damage, in particular, small dents on the tubes and any scratches, especially on the mounting surface of the base plate.

8. Retain the cold head box and packing material to use in the future if you need to store the cold head or ship it to Cryomech, Inc.

4.3.3 Helium flex lines

1. See weights and dimensions of the flex lines in Section 5 to ensure components can be moved safely.

2. Inspect the helium flex lines for any signs of damage.

4.3.4 Cold head motor cord

Inspect the cord for any signs of damage.

4.3.5 Tool kit

Cryomech, Inc. systems are shipped with a box of tools. The label on the box lists the contents included inside.

4.3.6 Helium compressor package

1. The helium compressor package should not be tipped more than 5° at any time. See weights and dimensions of the helium compressor package in Section 5 to ensure components can be moved safely.

2. Inspect the helium compressor package for any signs of damage such as dents, scratches or any signs of oil leaks.

3. Check the pressure readings on both pressure gauges. If either gauge reads 5 psig (0.34 bar) lower than the static pressure listed in Section 5, contact Cryomech, Inc.

4.4 Returning a system to Cryomech, Inc.

1. Contact Cryomech, Inc. for a Return of Material Authorization (RMA) number, and for additional detailed instructions on how to properly return system components.

2. Repackage the system:

IMPORTANT

Use the original crate and packaging materials to minimize the likelihood of damage during shipping.

a) Using the original packing material, rewrap the cold head and place in the original cold head box.

b) Place the helium compressor package on the pallet on top of sufficient vibration dampening material to prevent the wheels from touching the pallet.

c) Strap the helium compressor package to the pallet, making certain that there is sufficient insulating material between the helium compressor package and the straps so the straps will not scrape any paint off the helium compressor package.

d) Use protective wrap to secure the helium flex lines and place them in the container.

3. Include shipping labels on the crate showing which side is up and make clear that the shipment is fragile.

4. Cryomech, Inc. highly recommends using “tip and tell” indicators. These indicators are helpful in determining whether your package was handled properly. Replace used “tip and tell” indicators with new ones (three per crate).

5. When the system has been completely packaged, contact Cryomech, Inc. for shipping instructions.

Section 5

Specifications

Section 5: Specifications

5.1. Intended use of equipment

The Cryomech cryocooler system is used for cooling to cryogenic temperatures. Current clients include research laboratories and universities as well as manufacturers of MRI and NMR superconducting magnets as well as other types of equipment that require cryogenic temperatures.

5.2. Technical specifications

Following are the detailed technical specifications for the cold head and the compressor package.

CAUTION

Operation of the cryocooler in any situation that does not meet the specifications in this section will void the warranty. If you plan to operate the system outside any of the specified conditions, contact Cryomech, Inc.

5.2.1. System weight and dimensions

Parameter	Value	
Cold head weight	57 lb	25.9 kg
Cold head dimensions	See cold head drawing Figure 5-8	
Water cooled helium compressor package weight	420 lb	190.5 kg
Water cooled helium compressor package dimensions (L x W x H)	24 in x 24 in x 31 in	61 cm x 61 cm x 78.7 cm
High pressure flex line weight	60 lb	27.22 kg
High pressure flex line dimensions	1in ID, Low Noise, 3/4 in Aeroquip both ends, 66 ft	
Low pressure flex line weight	60 lb	27.22 kg
Low pressure flex line dimensions	1in ID, Low Noise, 3/4 in Aeroquip both ends, 66 ft	

5.2.2. Cooling water specifications

Parameter	Value	
Cooling water	See Figure 5-1	
Maximum inlet pressure	110 psig	7.6 bar
Alkalinity	5.8 < pH < 8.0	
Calcium carbonate	Concentration < 80 PPM	

INFORMATION

Cooling water temperature and flow rate should fall within the shaded area of Figure 5-1 to maintain optimal performance. Operating outside of the shaded area may result in system entering a warning or error state.

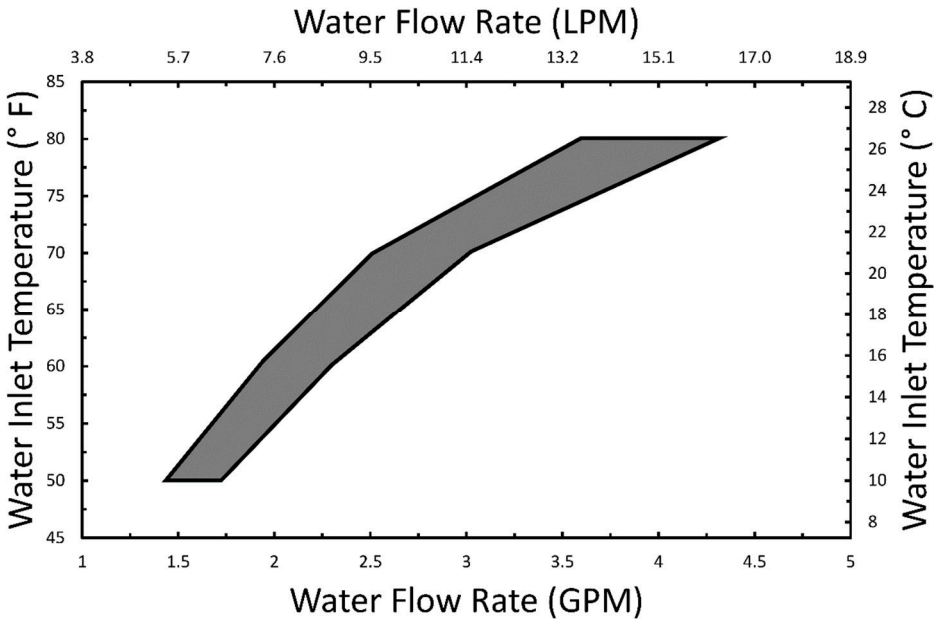


Figure 5-1: Cooling water requirement*

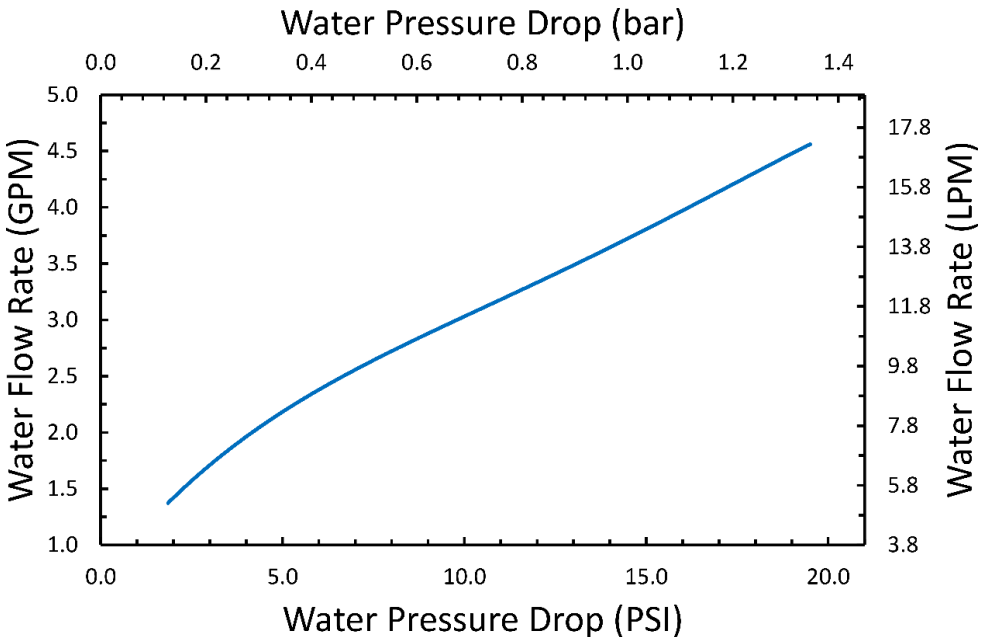


Figure 5-2: Cooling water pressure drop*

*When using a 50-50% mixture of ethylene glycol and water increase the flow rate 10%. Pressure drop values will increase 40%.

CAUTION

The use of DI (De-Ionized) water is prohibited. DI water causes damage to the compressor package and will not be covered under warranty.

5.2.3. Electrical specifications

Electrical Rating	Input Power (kW)*		Maximum Current (A)*	Dedicated Circuit Breaker (A)
	Maximum	Steady State		
440/480 VAC 60 Hz	14.5	13.0	21	25
380/415 VAC 50 Hz	13.5	11.4	22	25

*Measurement tolerance is +/- 5%

5.2.4. Operating parameters

Parameter	Value	
Ambient temperature range*	45 °F to 100 °F	7 °C to 38 °C
System helium pressure	220 ± 5 psig @ 60 Hz 260 ± 5 psig @ 50 Hz	15.2 ± .34 bar @ 60 Hz 17.9 ± .34 bar @ 50 Hz
Acceptable location	Indoors only	
Maximum altitude for use	6560 ft	2000 m
Environment	Pollution Degree 2	
Installation	Category II	
Maximum relative humidity	80% for T< 88 °F Decreasing linearly to 50% at 104 °F	80% for T< 31 °C Decreasing linearly to 50% at 40 °C
Water cooled maximum sound level	76 dBA at 1 meter	
Cold head maximum load 1st stage heat exchanger	22 lb	10 kg
Cold head maximum load 2nd stage heat exchanger (if applicable):	11 lb	5 kg

*The compressor package is designed to operate in an ambient temperature range from 45 °F to 100 °F (7 °C to 38 °C). If the temperature is below 45 °F (7 °C), the increased viscosity of the oil could prevent start-up and/or cause poor lubrication. Operation above 100 °F (38 °C) will cause overheating and subsequent problems. If a unit must be subjected to either extreme, written authorization from Cryomech, Inc. must be granted.

5.2.5. Safety Devices

Several safety devices are located inside the helium compressor package and on the cold head. They are monitored by the helium compressor package's control panel to maintain a safe working environment and protect the helium compressor package and cold head from developing extreme conditions that can cause damage, see Section 7 for further detail. The safety devices listed below are not monitored by the helium compressor package's control panel.

High-Pressure Atmospheric Relief Valve

The helium compressor package high-pressure atmospheric relief valve is set at 420 ± 5 psig (29 ± .34 bar). At pressures above 420 psig (29 bar) the relief valve will open automatically and relieve pressure to the atmosphere.

Internal Motor Overload Switch

An internal motor overload switch, located inside the helium compressor package module, protects the system by sensing excessive current draw and temperature. This switch automatically resets itself after the helium compressor package module cools to an acceptable level.

Cold Head High Pressure Relief Valve

The cold head high pressure atmospheric relief valve is set at 425 ± 5 psig (29.3 ± .34 bar). At pressures above 425 psig (29.3 bar) the valve will open automatically and relieve pressure to the atmosphere.

5.3. Description of helium compressor package

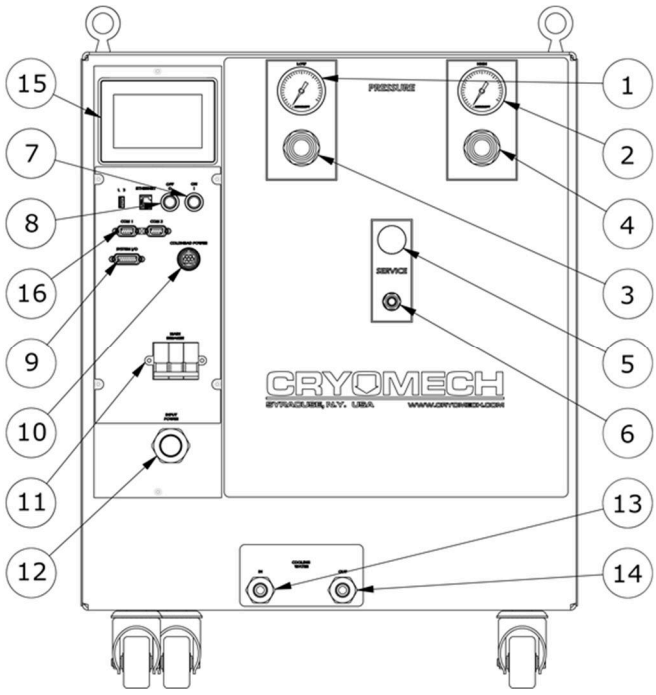


Figure 5-3: Front panel of the water cooled helium compressor package

5.3.1. Front panel interfaces

This section describes the function of the interfaces on the CPA1114 helium compressor package.

1. **Low Pressure Gauge**
   The pressure of the helium gas being returned to the helium compressor package from the cold head.
2. **High Pressure Gauge**
   The pressure of the helium gas being sent to the cold head from the helium compressor package.
3. **Low Pressure Aeroquip**
   Flex line connection for gas returning from the cold head to the suction side of the helium compressor package.
4. **High Pressure Aeroquip**
   Flex line connection for gas going to the cold head from the discharge side of the helium compressor package.
5. **Service Valve**
   Regulates the gas flow through the service access Aeroquip.
6. **Service Access**
   Helium compressor package connection used to add or remove helium.
7. **Helium Compressor Package ON Button**
   A mechanical button used to start the system.
8. **Helium Compressor Package OFF Button**
   A mechanical button used to stop the system.
9. **System I/O Port**
   A DB15 female socket that can be used to remotely control the helium compressor package and monitor a limited number of its parameters.
10. **Cold Head Motor Connector**
    The electrical connection for the cold head motor cord, from the helium compressor package to the cold head or remote motor assembly.
11. **Main Circuit Breaker**
    Provides over-current protection for the helium compressor package and functions as a power disconnect.
12. **Power Cord**
    Electrical connection that supplies power to the system.
13. **Cooling Water Inlet Connection**
    A 3/8 FPT (Female National Pipe Thread) fitting used to provide coolant to the helium compressor package.
14. **Cooling Water Outlet Connection**
    A 3/8 FPT (Female National Pipe Thread) fitting used as an exit for the coolant from the helium compressor package.
15. **Touch Screen**
    Displays system status, warnings, error messages, operation data, system information and allows users to interact with the helium compressor package.
16. **COM Ports**
    Used for remote monitoring and control of the system via SMDP or Modbus RTU protocol.

5.3.2. Control panel description

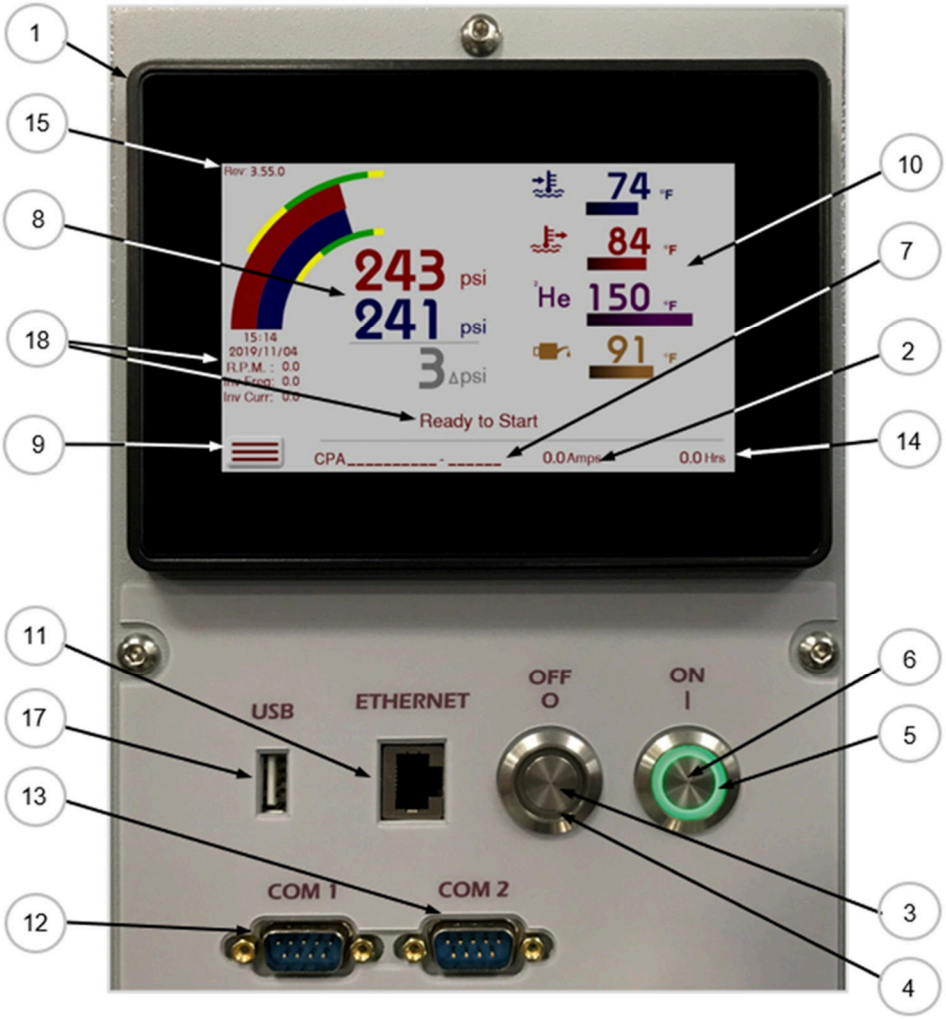


Figure 5-4: Control panel of the helium compressor package

1. **Touch Screen**
   Displays system status, warnings, error messages, operation data, system information and allows users to interact with the helium compressor package.
2. **Current Draw**
   Displays in amps, the current draw of the system.
3. **Helium Compressor Package OFF Button**
   Mechanical interface used to stop the system.
4. **Helium Compressor Package OFF Indicator Light**
   Illuminates red when the system is off, flashes red with the system is in an error state.
5. **Helium Compressor Package ON Button**
   Mechanical interface used to start the system.
6. **Helium Compressor Package ON Indicator Light**
   Illuminated green when the system is running.
7. **Helium Compressor Package Serial Number**
   Unique number used for tracking purposes and to identify the helium compressor package model.

8. **High and Low Pressures**
   Displays a running average of the suction and discharge pressure.
9. **Menu Button**
   Allows navigation to other screens, additional controls, and system information.
10. **Operating Temperatures**
    Displays the coolant in and out temperature, helium temperature and oil temperature.
11. **RJ45 Ethernet Port**
    Used for remote monitoring and control of the system via Modbus TCP protocol.
12. **RS232 Port**
    Used for remote monitoring and control of the system via SMDP or Modbus RTU protocol.
13. **RS485 Port**
    Used for remote monitoring and control of the system via SMDP or Modbus RTU protocol.
14. **Running hours**
    Total number of hours the helium compressor package has run, this does not include idle time.
15. **Software Version**
    Unique number used for tracking changes to the PLC program.
16. **System I/O Port**
    Used to remotely control the helium compressor package and monitor a limited number of its parameters.
17. **USB Port**
    Used to update the helium compressor package software and download recorded data from the PLC.
18. **User Feedback Text**
    Displays the status of the helium compressor package including errors, warnings, cold head rpm, and date and time.

5.3.3. System I/O

The digital I/O DB15 female connector is described in this section. Enabling the inputs require a minimum differential of 12 Vdc between the input signal pins and the input reference pin. The input supply voltage range is 20.4 to 28.8 Vdc. Input voltage must drop below 5 Vdc and 1 mA to be disabled. The state of input pins is changed on the rising edge of the signal. The pulse width for an input signal must be a minimum 200ms to guarantee the signal is detected. The inputs are ESD protected and have an impedance of approximately 3.3K Ohm.

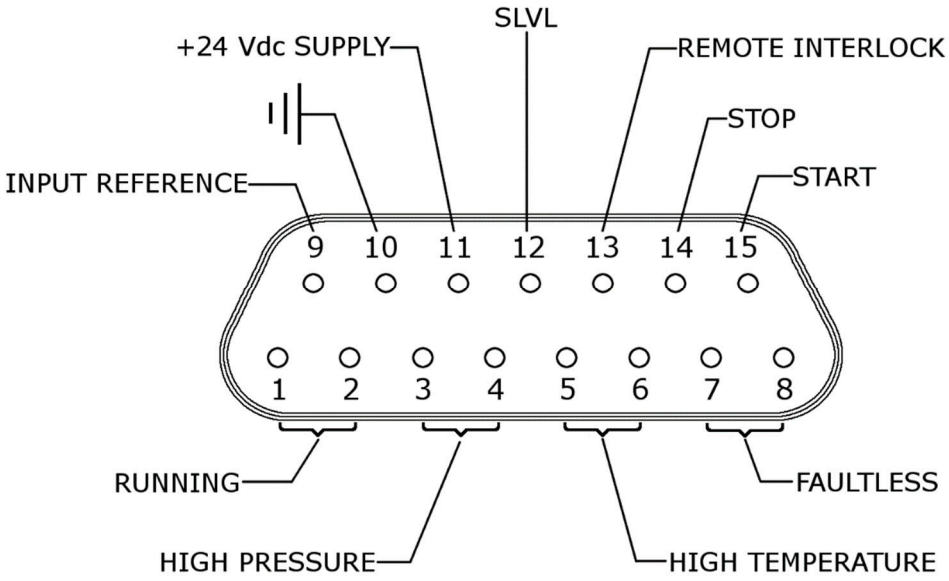


Figure 5-5: 15 Pin connector diagram

Note: In this manual, the System I/O labels have but updated to better reflect functionality. The overall operation of the System I/O has not changed.

Inputs

SLVL (Pin 12)

When SLVL is enabled, the Start pin must remain enabled for the system to run. Enabling the Start pin will put the compressor into Starting mode, then enter Running mode. If the Start pin is disabled, the system will enter Stopping mode. If the Compressor OFF Button, Screen OFF Button, Modbus or SMDP OFF command, Remote Interlock pin or Stop pin are enabled, while the system is running the system will enter Stopping mode. If the system is off when the Remote Interlock pin or Stop pin are enabled the system will not start. When SLVL is enabled, the Compressor ON Button, Screen ON Button, Modbus and SMDP start commands are disabled, enabling the System I/O Start pin is the only way the system will enter Starting mode.

Remote Interlock (Pin 13)

When the Remote Interlock is enabled the system cannot be started. Enabled the Remote Interlock, disables the Compressor ON Button, Screen ON Button, System I/O Start, Modbus and SMDP start commands. If the system is running when the Remote Interlock is enabled, the system will enter Stopping mode.

Stop (Pin 14)

When the Stop is enabled, the system cannot be started. Activating Stop disables the Compressor ON Button, Screen ON Button, System I/O Start, Modbus and SMDP start commands. If the system is running when Stop is enabled, the system will enter Stopping mode.

Start (Pin 15)

When Start is enabled, the compressor will enter Startup mode then Running mode. If there is an off command coming from any input; Compressor OFF Button, Screen OFF Button, System I/O Stop, System I/O Interlock, Modbus or SMDP stop command, or an error, the system will not start.

Outputs

Running (Pins 1-2)

When the system is in Running mode, and operating properly, the Running relay between pins 1 and 2 will closed.

High Pressure (Pins 3-4)

If the discharge pressure of the compressor package exceeds the high pressure limit the High-Pressure relay between pins 3 and 4 will close.

High Temperature (Pins 5-6)

If the helium gas of the compressor package exceeds the high temperature limit the High temperature relay between pins 5 and 6 will close.

Faultless (Pins 7-8)

If the system is powered on and in a running or idle state and the system is experiencing no warnings or errors the relay between pins 7 and 8 will close.

Voltage supply

Pin 11 is +24 Vdc source referenced to the ground, pin 10. The maximum current available is 40 mA. This supply can be used to power the inputs. The supply may also be used for other purposes provided the current limit is not exceeded.

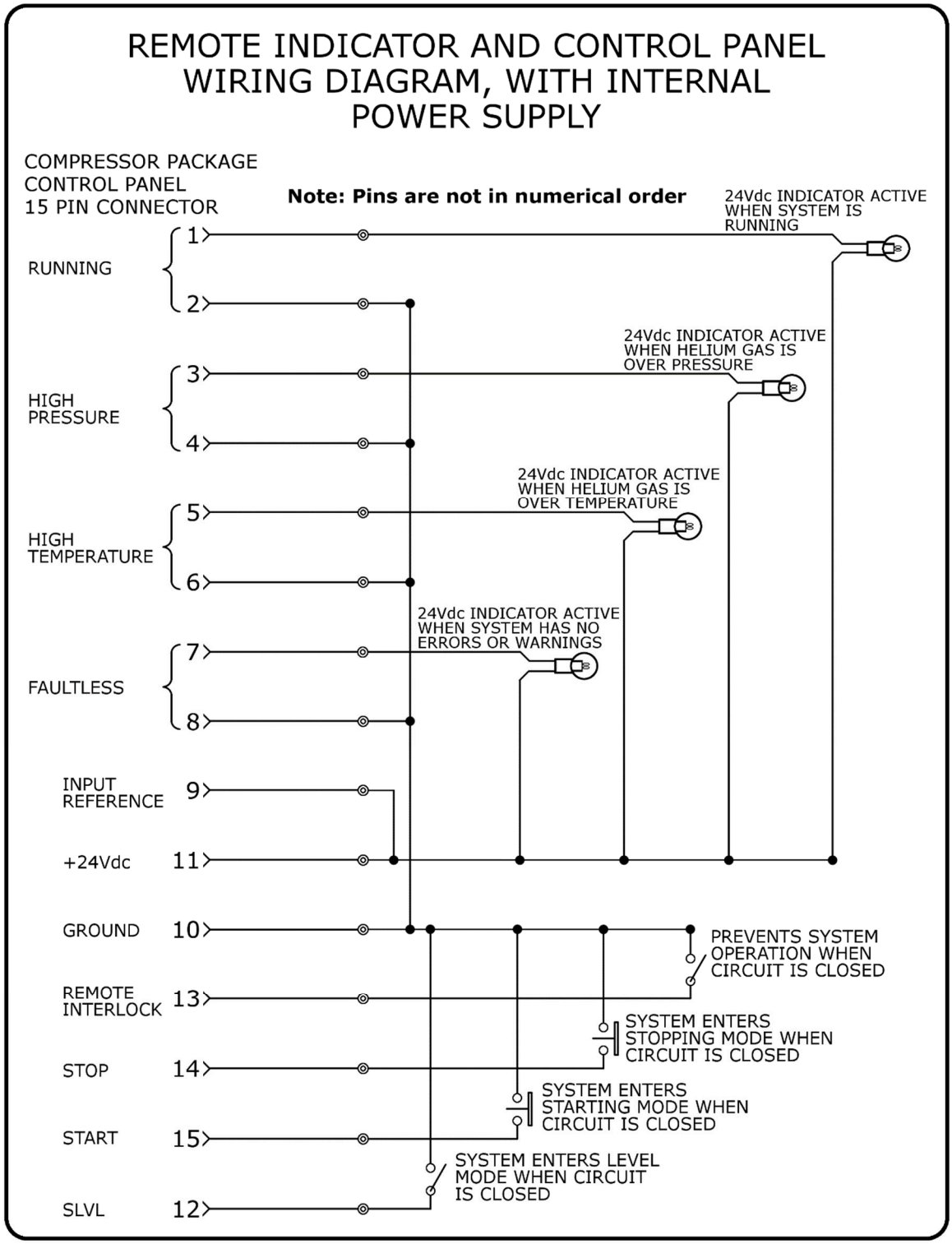


Figure 5-6: System I/O wiring diagram

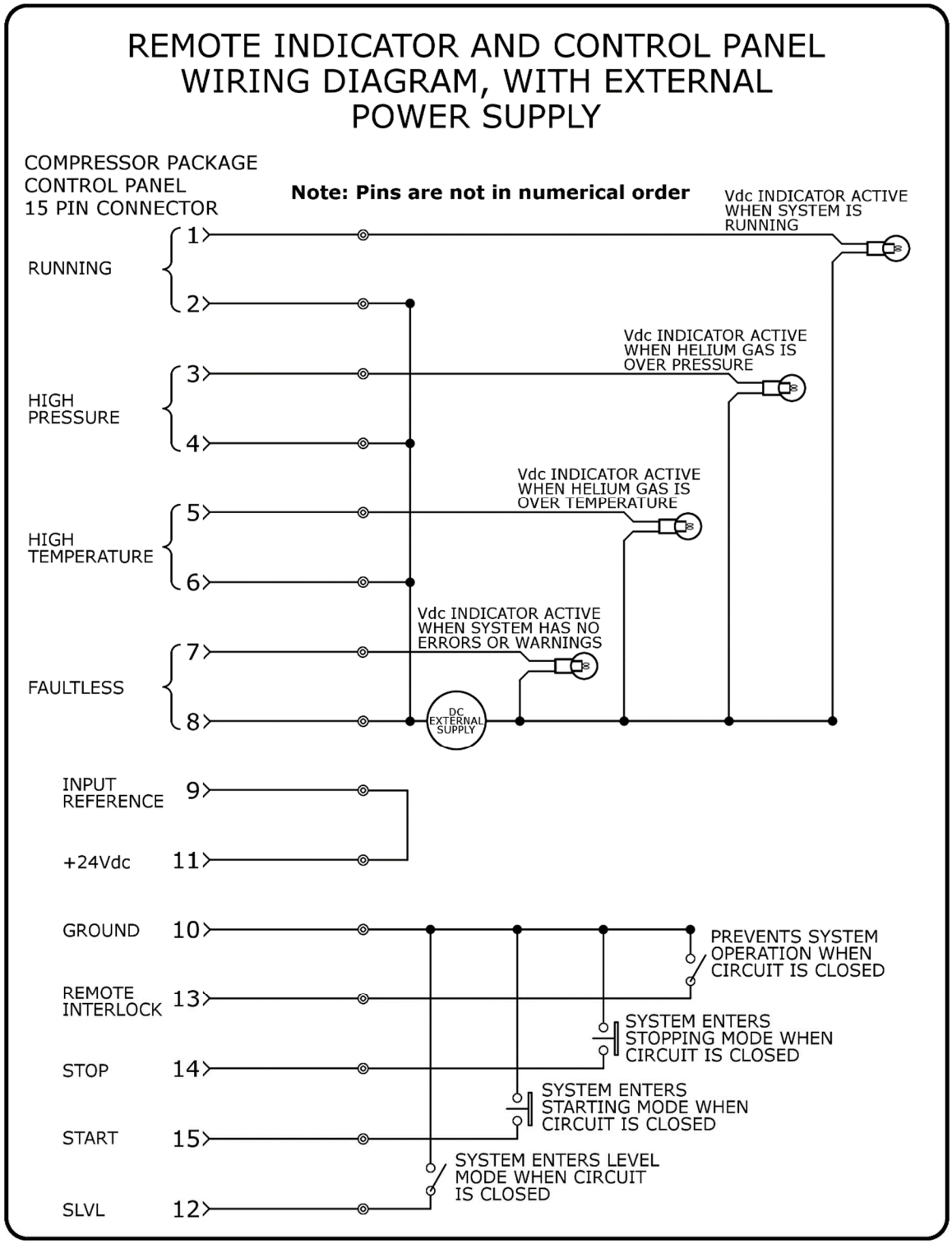


Figure 5-7: System I/O wiring diagram – external supply

5.4. Cold head drawing

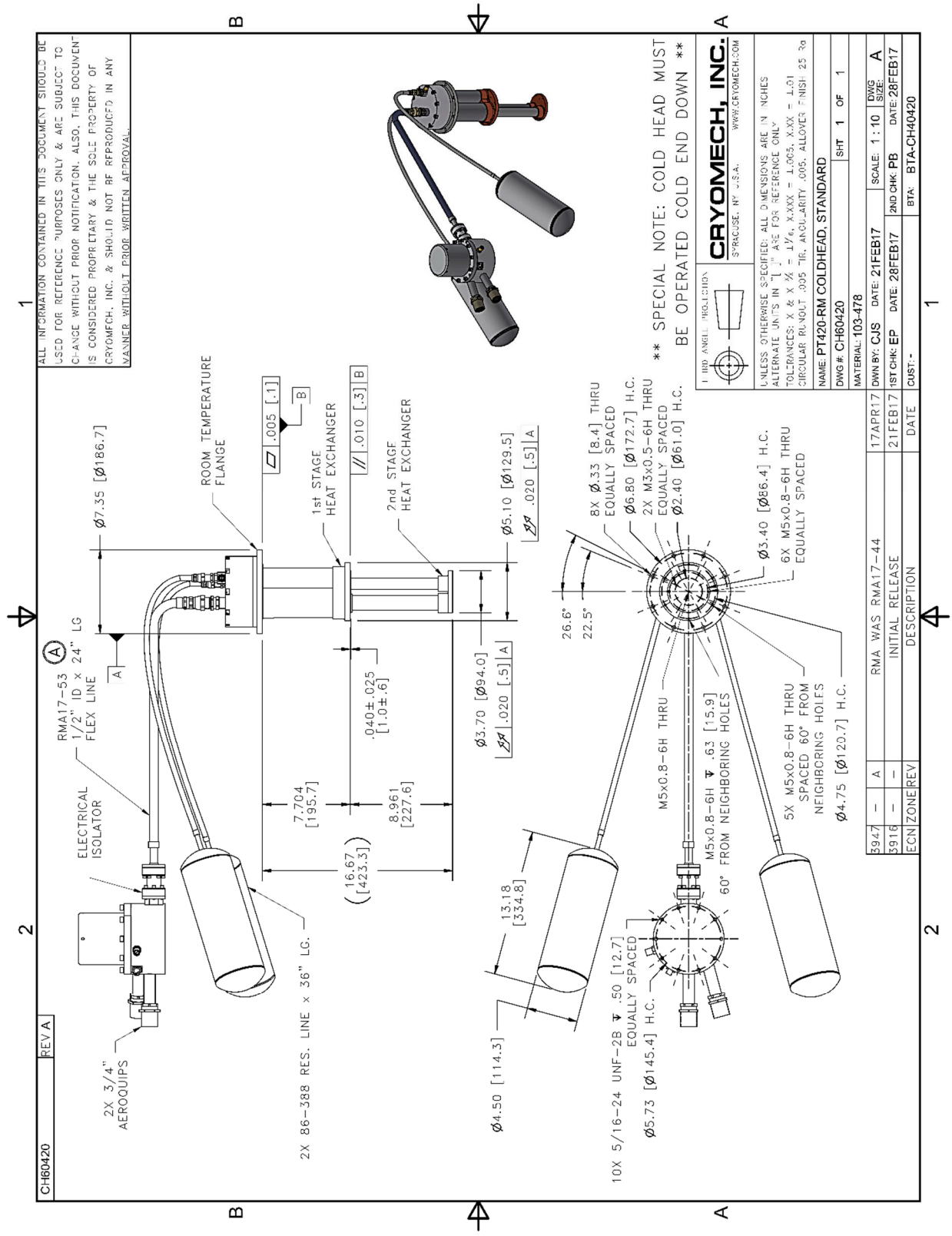


Figure 5-8: Cold Head Drawing

Section 6

Installation

Section 6: Installation

Review the entire section on the installation before installing the system.

6.1 Cold head installation

CAUTION
Failure to follow these installation guidelines will void the warranty.

CAUTION
Special care must be taken when handling the cold head. The warranty will not cover dented or bent tubes.

CAUTION
Operating the cold head in a magnetic field of greater than 500 gauss will void the warranty.

1. Inspect the mounting surface of the base plate for any scratches in the sealing area. Scratches or dents on the sealing surface will cause a leak from the ambient into the vacuum vessel.
2. Inspect the interface surfaces of the copper heat exchanger for scratches or dents before attaching anything to it. Scratches or dents will prevent good thermal contact between the heat exchanger and the object(s) attached to it.
   - Please notify Cryomech, Inc. if scratches or dents are observed on any of the above-mentioned surfaces.
3. Make certain the flat gaskets are present and properly seated in the ends of the Aeroquip connectors.

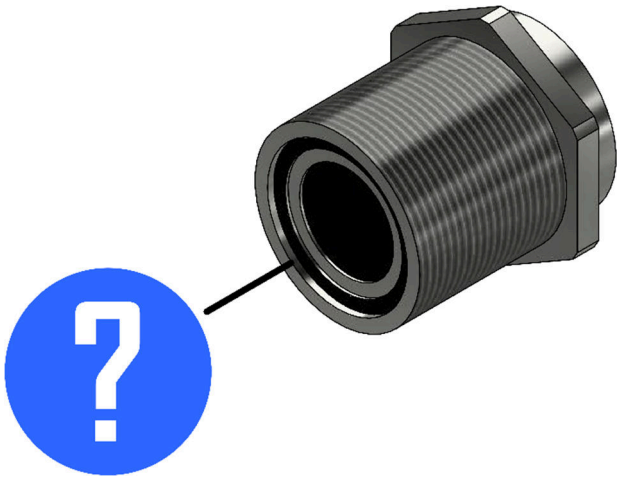
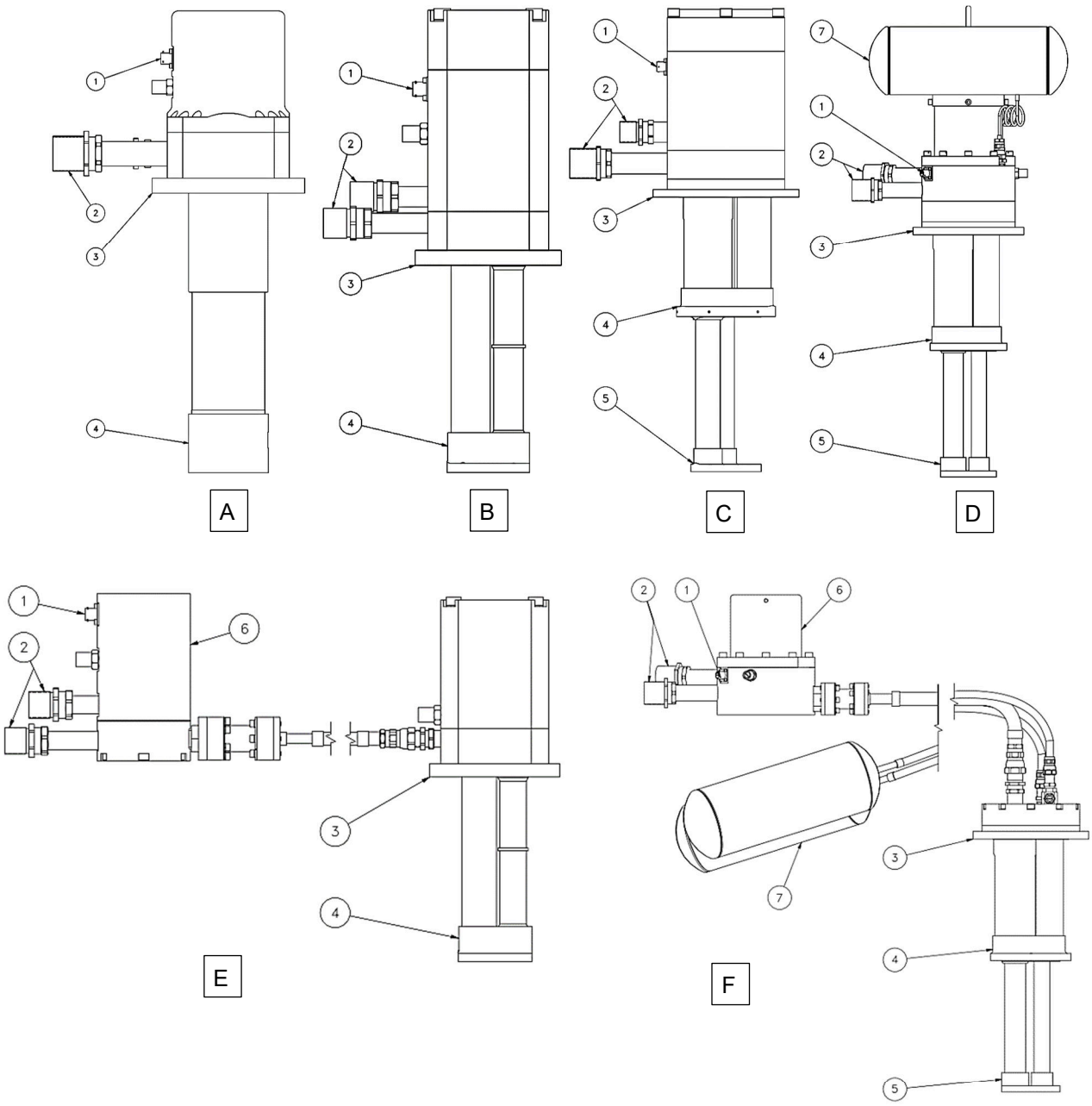


Figure 6-1: Male Aeroquip gasket

6.1.1 Inspect the cold head



Item #	Description
A	GM
B	Single Stage PT
C	2 Stage PT
D	2 Stage PT w/ Reservoir Volume
E	Single Stage Remote Motor PT
F	2 Stage Remote Motor PT

Item #	Description
1	Cold Head Motor Cord Connector
2	Aeroquip Connectors
3	Base Plate/Room Temperature Flange
4	First Stage Heat Exchanger
5	Second Stage Heat Exchanger
6	Remote Motor Assembly
7	External Reservoir

Figure 6-2: Outline drawing of standard cold heads

6.1.2 Initial preparation of the cold head

1. Clean the mating surfaces of the base plate and heat exchanger(s) with isopropyl alcohol.
2. Apply a small amount of vacuum grease to the mating flange O-ring (not supplied) and insert in the O-ring groove.
3. Join the mating flange of the vacuum chamber onto the cold head. Secure the flange to the base plate with the proper number of screws. Be certain to tighten the screws evenly around the mating surface.
4. Prepare the desired cold mating surface(s) and, applying a small amount of Apiezon N grease or pure Indium foil on all cold mating surfaces, secure to the heat exchanger(s) using brass screws.

CAUTION

Heating any part of the cold head above 325 K (125 ºF, 52 ºC) will cause damage and void the warranty.

CAUTION

With the cold head oriented vertically, the maximum allowable load cannot exceed the value specified in Section 5. Excessive loading will damage the cold head. This type of damage is not covered by the warranty. If your application requires a load heavier than the weight specified in Section 5, prior written authorization from Cryomech, Inc. must be granted.

CAUTION

Mounting the cold head in an orientation other than vertical will void the warranty without prior written authorization from Cryomech, Inc.

CAUTION

Alterations to the cryocooler are prohibited without prior written authorization from Cryomech, Inc. Modifications without prior written approval will void the warranty.

6.1.3 Optimal installation of temperature sensors (optional)

There are three critical areas if the application requires the use of temperature sensors:

1. Proper mounting of the sensor(s) on the heat exchanger(s).
2. Proper joining of sensor lead wires and connecting wires.
3. Thermal anchoring of the lead wires to the heat exchanger(s).

Please contact Cryomech, Inc. if you have poor performance or need more specific details on installation of the sensors.

6.1.4 Installation of heaters (optional)

There are three critical areas if the application requires the use of heaters:

1. Proper mounting of the heater(s) on the heat exchanger(s).
2. Proper joining of heater lead wires and connecting wires.
3. Thermal anchoring of the lead wires to the heat exchanger(s).

Please contact Cryomech, Inc. if you have poor performance or need more specific details on installation of the heaters.

6.1.5 Mounting the radiation shield (optional)

1. Clean the radiation shield with isopropyl alcohol.
2. Apply a small amount of Apiezon N grease or pure Indium foil on the shield mating surface. Attach the radiation shield with brass screws. Tighten it evenly to the first stage heat exchanger.

6.1.6 Connect the remote motor assembly's flex line (if applicable)

1. Remove the dust plug from the Aeroquip fitting attached to the end of the remote motor's flexible line and place in the tool kit for future use.
2. Remove the dust cap from the cold head's mating Aeroquip fitting (labeled "Motor") and place in the tool kit for future use.
3. Make certain the flat gasket is present and properly seated in the cold head's Aeroquip fitting. Reference Figure 6-1
4. With a dry, clean lint-free cloth remove any visible particles from the ends of the Aeroquip couplings.

IMPORTANT

When connecting or disconnecting Aeroquips, Cryomech, Inc. recommends using a small amount of dry Teflon spray lubricant. Follow the manufacturer's instructions for its use. Do not spray lubricant onto sealing surfaces of the Aeroquips. First, hand tighten Aeroquips, then spray small amount lubricant on the threads of the male Aeroquip. See Figure 6-3.

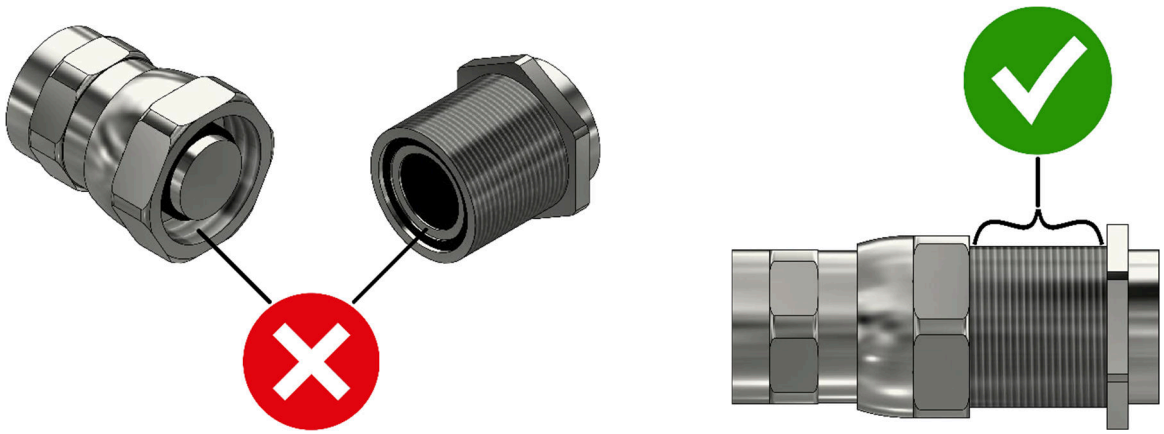


Figure 6-3: Dry Teflon spray lubricant guidelines

CAUTION

Follow this procedure carefully when connecting and disconnecting the helium flex lines. Failure to follow the procedure will void the warranty.

CAUTION

Do not allow the flex lines to contact any corrosives or commercial cleaning agents. Flex line damage caused by exposure to corrosives or commercial cleaning agents will not be covered by the warranty.

CAUTION

Applying extreme bend stress to a flex line will cause permanent damage. Bending a flex line to less than a 12 in (30 cm) radius will void the warranty.

5. Connect the Aeroquip fittings until a positive stop is felt. See Figure 6-3.

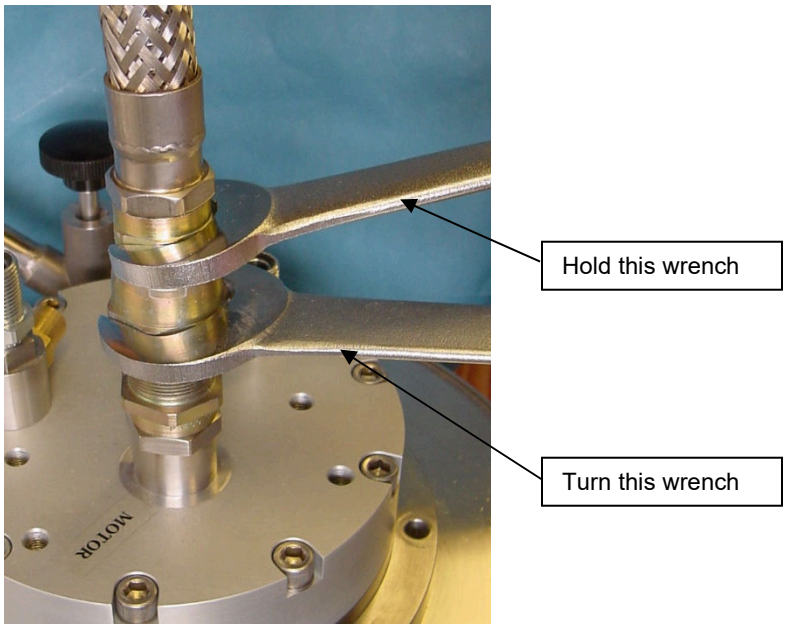


Figure 6-4: Connecting the remote motor assembly's flex line to the cold head

6.1.7 Attaching the external reservoirs (if applicable)

1. Remove the dust plugs from the external reservoirs' Aeroquip fittings and place in the tool kit for future use.
2. Remove the dust caps from the cold head's Aeroquip fittings and place in the tool kit for future use.
3. Make certain the flat gaskets are present and properly seated in the cold head's Aeroquip fittings. Reference Figure 6-1
4. With a dry, clean lint-free cloth remove any visible particles from the ends of the Aeroquip couplings.

5. Connect the Aeroquip fittings until a positive stop is felt. The external reservoirs are identical and therefore, interchangeable. A dry Teflon spray lubricant is recommended, See Figure 6-3.

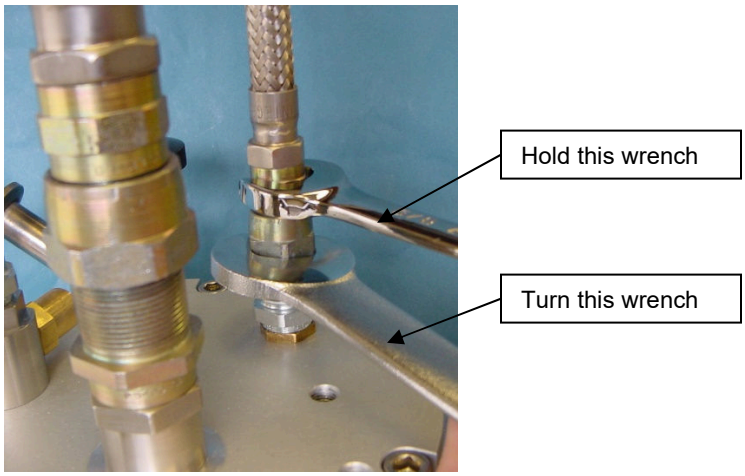


Figure 6-5: Connecting the external reservoir's flex line to the cold head

6.2 Compressor installation

CAUTION

Failure to follow these installation guidelines could result in voiding the warranty.

6.2.1 Prepare the helium compressor package location

- Confirm that the physical space containing the helium compressor package packages has an ambient temperature range of 45 °F to 100 °F (7 °C to 38 °C).
- Place the helium compressor packages in a level position. For the helium compressor package to operate under optimal conditions, it must be oriented within 5° of being level.

WARNING

The helium compressor package must be positioned to provide easy access to the front panel mounted circuit breaker.

- Locate the helium compressor package so that it may be easily repositioned to allow for sufficient space change the adsorber. If the helium compressor package cannot be easily repositioned, leave approximately two feet (0.6 m) clearance above, to the left and right of each of them.
- In addition to the above requirements, CP100 series air cooled models require two feet (0.6 m) of clearance on each side of the unit to ensure proper air flow. CPA2800 series air cooled models require two feet (0.6 m) of clearance around all four sides and three feet (1 m) of clearance above for proper air flow.

6.2.2 Connect the water lines to the helium compressor package

CAUTION

Cooling water must meet the requirements in Section 5. Introduction of cooling water that does not meet the specifications in Section 5, even for cleaning purposes, it will void the warranty.

CAUTION

Do not apply heat to the cooling water inlet and outlet connectors located on the front panel of the helium compressor package.

1. Make sure that the cooling water supply is turned off.
2. Apply Teflon tape or pipe sealant to the threads on the male pipe thread (MPT) to connect to the compressor's cooling water inlet connection and the cooling water outlet connection. (Fittings included in the tool kit are optional)
   - Note: Compressors require 3/8 MPT (3/8 in Male National Pipe Thread) fittings.
3. Attach the fittings to the helium compressor package package's cooling water inlet and outlet connections. Use a wrench to keep the bulkhead from turning.
4. Attach the supply water line to the COOLING WATER IN fitting and the return water line to the COOLING WATER OUT fitting.
5. Turn the cooling water supply on and check for leaks.
6. Make certain the cooling water flow rate and inlet temperature meets the requirements in Section 5.

6.2.3 Connect the helium compressor package to the main power

CAUTION

The mains supply voltage to the helium compressor package must be within 10% of the nominal voltage specification in Section 5. Operating the helium compressor package outside the voltage specification will void the warranty.

1. Each helium compressor package must be connected to a dedicated circuit breaker. The breaker must be mounted near the helium compressor package, within easy reach of the operator and must be marked as the disconnecting device for the system. Specifications for circuit breakers vary according to the system's operating voltage. See the electrical specification tables in Section 5 for more information.

WARNING

Follow all local electrical codes and guidelines when during installation.

2. Make sure that the dedicated circuit breaker is turned off.
3. The ground (or earth) wire in the power cord is either green or green/yellow stripe. Connect the ground wire in the power cord to the ground (or earth) connector in the breaker panel. It is important not to disable this wire.
4. Connect the remaining power cord wires to the corresponding lugs on the dedicated breaker. The order of the wires is not important at this time - correct order will be determined in Section 7.

WARNING

One lead of the helium compressor package is grounded. Never bypass this ground or attach the helium compressor package to an ungrounded circuit. A dangerous electrical hazard will develop.

6.2.4 Connect the helium flex lines to the helium compressor package and the cold head

CAUTION

Follow this procedure carefully when connecting and disconnecting the helium flex lines. Failure to follow the procedure will void the warranty.

CAUTION

Do not allow the flex lines to contact any corrosives or commercial cleaning agents. Flex line damage caused by exposure to corrosives or commercial cleaning agents will not be covered by the warranty.

CAUTION

Applying extreme bend stress to a flex line will cause permanent damage. Bending a flex line to less than a 12 in (30 cm) radius will void the warranty.

1. Remove the dust plug from the Aeroquip fitting attached to the end of flexible line and place in the tool kit for future use.
2. Remove the dust caps from the cold head's and helium compressor package packages mating Aeroquip fitting and place in the tool kit for future use.
3. Make certain the flat gasket is present and properly seated in the Aeroquip fittings on the cold head and helium compressor package, See Figure 6-1.
4. With a dry, clean, lint-free, cloth, remove any visible particles from the ends of the Aeroquip couplings.

5. Connect a helium flex line marked LOW to the low pressure port on the helium compressor package front panel. Connect a helium flex line marked HIGH to the high pressure port on the helium compressor package front panel. When attaching the Aeroquip to the mating connector, make sure the threads are in alignment before you tighten the connector. Tighten the connector until a positive stop is felt. A dry Teflon spray is recommended. See Figure 6-3

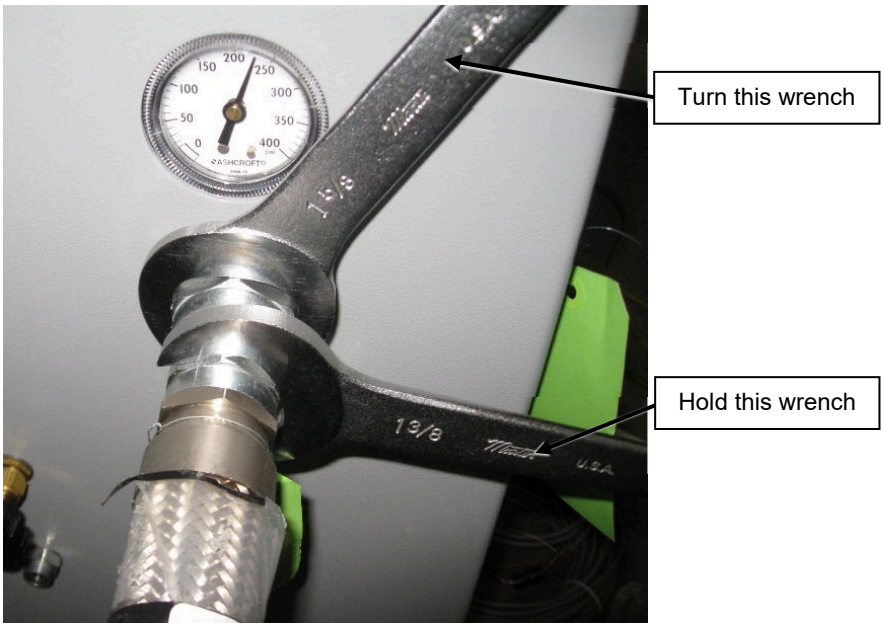


Figure 6-6: Connecting the flex lines to a helium compressor package

6. Connect helium flex line marked LOW to the low pressure port on the to the cold head or remote motor. Connect a helium flex line marked HIGH to the high pressure port on the cold head or remote motor. Tighten the connector until a positive stop is felt. A dry Teflon spray is recommended. See Figure 6-3

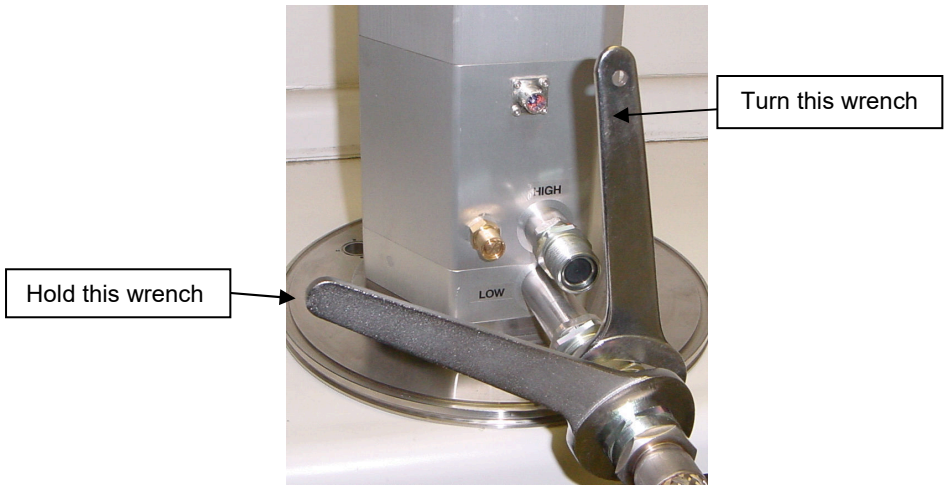


Figure 6-7: Connecting the low-pressure flex line to the low-pressure port on a cold head

6.2.5 Connect the cold head motor cord

1. Connect the cold head motor cord from the helium compressor package to the cold head, making sure both connectors are properly coupled.

6.2.6 Remove excess helium from the helium compressor package

1. Turn the circuit breakers on to provide power to the helium compressor package.
2. Navigate through the PLC to find the helium pressure screen.
3. If the pressures exceed what is indicated in Section 5, make sure that the service valve is closed.
4. Attach the ¼ inch service Aeroquip to the service access.
5. Slowly open the service valve to reduce the static charge. Do not vent more than 5 psi (0.34 bar) of gas per minute.
6. Observe the PLC pressure reading. Once the pressure reaches the static charge indicated in Section 5, close the service valve.
7. Remove the ¼ inch service Aeroquip from the service access.

Section 7

Operation

Section 7: Operation

7.1 Starting the system

When the installation is complete, make the following checks before starting the system. Carefully follow the steps outlined in this section to ensure proper startup and operation of the system.

7.1.1 Checks before operating

1. The dedicated circuit breaker is switched on.
2. The cooling water meets requirements of flow rate and temperature range, specified in Section 5.
3. Make sure all Aeroquip couplings are securely fastened and the helium flex lines are oriented correctly (High port to High port, Low port to Low port).
4. The helium pressure is as follows:
   - Gauges should read as specified in Section 5 of this manual.
   - If the pressure is too high, refer to Section 8 of this manual.
   - If the pressure is too low, refer to Section 8 of this manual
5. Make sure the cold head motor cord is connected to both the helium compressor package and the cold head.
6. Make sure the input power meets the specifications on the identification label.

CAUTION

A voltage deviation of more than 10% above or below the voltage rating can cause helium compressor motor overheating and possible failure. Indications of voltage operation outside that range will void the helium compressor warranty.

CAUTION

Operating the cold head in a magnetic field of greater than 500 gauss will void the warranty.

7.1.2 Startup procedure

1. Switch on power to the helium compressor package at the MAIN BREAKER.
   - The red OFF light will illuminate.
2. The compressor may be started by pushing the compressor ON button.

WARNING

The system operates in a closed loop cycle at pressures higher than 200 psig (12.8 bar) and at temperatures below 120 Kelvin.

7.2 Normal operation behavior

7.2.1 Normal pressures

On startup, a pressure differential should be noticed immediately between the high and low pressure gauges. This differential will decrease as the cold head temperature decreases.

The typical pressure differential is approximately 220 psi to 250 psi (15.2 bar to 17.2 bar) with a range of 5 psi to 10 psi (.34 bar to 0.7 bar) on the pressure gauge needles.

7.2.2 Normal sounds

When operating properly, the cryocooler will emit a rhythmic sound (squeak or chirp). This noise is an indication that there is proper flow of helium gas within the system

7.2.3 Cool down time

Refer to Section 5 for the cool down time of an unloaded cold head with no additional mass on the heat exchanger(s). If the cold head is exposed to a heat load or additional mass is attached, the cool down time will increase

7.2.4 Normal temperatures

Refer to Section 5 for the minimum no load temperature rating of the cold head.

This temperature will vary depending on the quality of the vacuum, the quality of the super insulation, and the heat load brought in by the instrumentation leads.

7.3 Use of the PLC

Status and error messages will be shown on the PLC screen. The status messages, error messages, and set points at which error conditions will occur are listed in Section 7.7.

The screen normally shown on the display is referred to as the run time screen. The messages described in Section 7.7 are displayed on the top line of the run time screen. The bottom line displays the number of hours the helium compressor package has run.

Please refer to the CPA User Guide (separate document) for detailed instructions on the capabilities of the helium compressor package display panel.

7.4 Shutdown procedure

Press the helium compressor package OFF button on the front panel of the helium compressor package, this will switch off the system. Switch off the front panel mounted circuit breaker to remove power from the helium compressor package's electrical panel

7.5 Recommended routine procedures

Cryomech, Inc. recommends keeping daily logs of system temperatures and pressures to detect changes in performance. This can include ambient air temperature, in and out cooling water temperatures, helium temperature and, high and low helium pressure fluctuation in performance could signify a problem that requires attention.

7.5.1 High and low pressure

Changes in the high and low pressure readings on the helium compressor package gauges are used for diagnosing several different types of problems. It is important to know whether changes are sudden or gradual and to know how the high and low pressures are changing relative to each other.

The high and low pressures should be monitored daily.

Cryomech, Inc. recommends that you maintain a regular record, at intervals that make sense for the way you use your system, of the high and low pressure readings

7.5.2 Cold head temperature

Temperature changes are the other key diagnostic.

If a temperature sensor is attached to the cold head heat exchanger(s), Cryomech, Inc. strongly recommends that you monitor the temperature(s) daily.

Cryomech, Inc. also recommends that you keep a regular record of the temperature(s) at intervals that make sense for the way you use your system.

7.5.3 Cooling water input and output (if applicable)

If possible, Cryomech, Inc. recommends keeping a regular record of the input and output cooling water temperatures and of the cooling water flow rate.

7.6 Disassembling the system for transport or storage

Use the following steps to prepare a Cryomech, Inc. cryocooler for eventual transport or storage.

1. Disconnect the main power to the system by switching the dedicated circuit breaker to the off position.
2. Disconnect the power cord from the external breaker panel. Coil up the power cord in preparation for transport or storage.
3. Disconnect the cold head motor cord from the cold head and from the helium compressor package. Coil the cold head motor cord in preparation for transport or storage.
4. Turn off the water supply at the source.
5. Using two wrenches disconnect the supply and return water lines from the inlet and outlet fittings by turning the fittings counterclockwise until they are released from the fittings. Store the connectors for transport or storage.
6. Drain the water from the helium compressor package. Blow remaining water out with compressed air.

IMPORTANT

4K PULSE TUBE SYSTEMS ONLY

Do not continue with the remaining steps until the cold head temperature has risen above 150 K. If the cold head is disconnected from the system while below 150 K, helium will expand within the cold head as it warms, escape through the pressure relief valve and require the user to recharge the system with helium before it would again operate properly.

7. Using the wrenches supplied in the tool kit, disconnect the high and low pressure flex lines from the helium compressor package by turning the Aeroquips counterclockwise with one wrench while holding the other wrench to prevent the flex line from twisting. See Figure 7-1.



Figure 7-1: Disconnecting flex lines from a helium compressor package

IMPORTANT

When connecting or disconnecting Aeroquips, Cryomech, Inc recommends using a small amount of dry Teflon spray lubricant. Follow the manufacturer's instructions for its use. Do not spray lubricant onto sealing surfaces of the Aeroquips, Hand tighten Aeroquips then spray small amount lubricant on the threads of the male Aeroquip. See Figure 7-2.

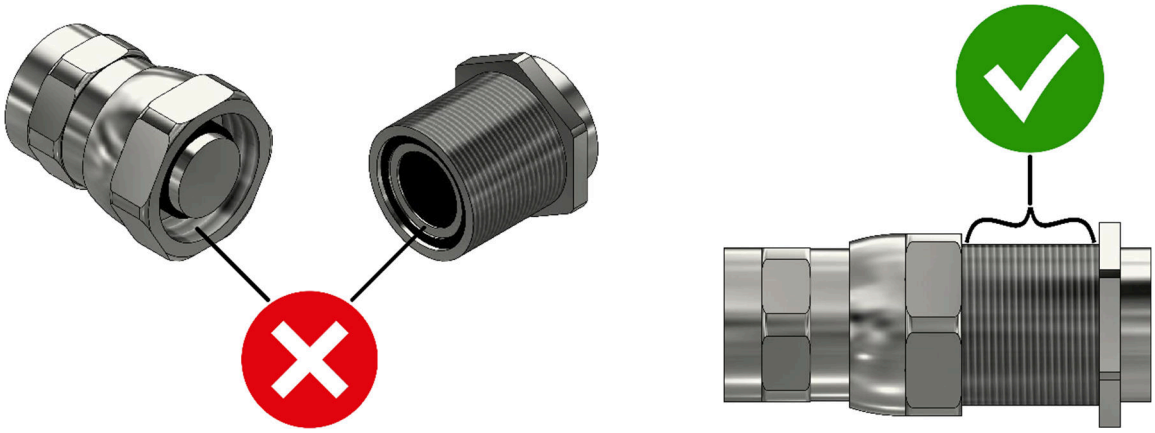


Figure 7-2: Dry Teflon spray lubricant guidelines

8. Disconnect the Aeroquips on the high and low pressure fittings on the cold head by turning the flexible line's Aeroquip counterclockwise with one wrench while holding the cold head's Aeroquip with the other wrench. See Figure 7-3.



Figure 7-3: Disconnecting flex lines from a cold head

9. Install the dust plugs (that originally came with the system and are stored in the tool kit) on the female Aeroquip couplings attached to the helium flex lines.

10. Install the dust caps on the male Aeroquip fittings located on the helium compressor package and the cold head.

11. If necessary, disconnect the cold head from its mounting surface, taking care not to damage any components. Once disconnected, place the cold head in the original packaging container.

WARNING

Do not touch the cold end of the cold head until it has warmed to room temperature. If there is frost on the cold head, it is too cold to touch.

12. Recoil the helium flex lines and prepare for transport or storage.

13. Assure that all components are stored in appropriate containers and location.

7.7 Troubleshooting

7.7.1 Helium compressor package will not start

SYMPTOM	System will not start.
POSSIBLE CAUSE	1) No power supplied to the helium compressor package. 2) Circuit breakers off.
REMEDY	1) Check the power supply to the system and verify that it meets the requirements outlined in Section 5. 2) Make certain the circuit breaker, located on the front panel of the helium compressor package, is on.

7.7.2 Helium compressor package starts, no pressure fluctuation

SYMPTOM	System starts, no bounce in the pressure gauges, no refrigeration.
POSSIBLE CAUSE	1) Cold head motor cord not connected to the cold head and/or to the helium compressor package. 2) Aeroquip connector(s) not completely tightened. 3) High and low pressure helium flex lines are reversed.
REMEDY	1) Turn off the helium compressor package and connect the cold head motor cord to the cold head and/or to the helium compressor package. 2) Tighten all Aeroquip connectors. 3) Verify that one of the helium flex lines connects the high pressure port on the helium compressor package to the high pressure port on the cold head and that the other helium flex line connects the low pressure ports.

7.7.3 System has shut itself down. Power switch light off

SYMPTOM	System has shut itself down.
POSSIBLE CAUSE	1) Circuit breaker tripped. 2) Interruption of the power supply to the helium compressor package. 3) “Compressor Error” message displayed on front panel screen.
REMEDY	1) Reset the circuit breaker on the front panel of the helium compressor package. See Section 5. 2) Check the power supply to the system and verify that it meets the requirements outlined in Section 5. 3) Refer to Section 7.7.5 for error message diagnostics.

7.7.4 Run time display screen description

Run time messages are displayed on the main display screen. The number of hours the system has operated is displayed in the bottom right hand corner.

Top line message on the display screen	Message description
“Idling....”	Front panel or remote interface turned off helium compressor package. All is well.
“Running Normally”	Compressor is running. All is well.
“Starting...”	Compressor ON request received, and helium compressor package will start within 10 seconds unless an error occurs, or a stop request is received.
“Helium Over Temp Recovery Delay…. WAIT x” (x is a number, counting down, in minutes & seconds. When x is zero compressor will start)	Compressor ON request received and helium compressor package will be able to be restarted in x amount of time unless an error occurs, or a stop request is received. This message will show when the helium compressor package has encountered certain errors that require a minimum off time, such as “Helium Temp. too high”
“You have had 6 errors in less than an hour: mm:ss” mm:ss indicates them minutes : seconds until the compressor can be reset.	The helium compressor package has entered a locked out state due to experiencing 6 errors within one hour. The helium compressor package cannot be restarted until the error lockout has been reset. You must select the lockout menu and reset the number of lockouts. In doing this you must acknowledge the relevant errors and therefore take the necessary steps to prevent these errors from reoccurring. Resetting the lockout status will enforce a 30 min cool down period before the helium compressor package can be restarted to prevent the short cycling of compressor module. The lockout contributors are noted in Section 7.6.5.

7.7.5 Error diagnostics on display screen

Errors will cause the helium compressor package to stop. Errors are displayed on the bottom line of the display screen. If more than one of the errors below is present, only the highest priority one will be displayed. The table lists errors from highest to lowest priority. If the error is a lockout contributor, it will be noted in the Explanation column.

Once the error condition has cleared, the helium compressor package can be restarted by pressing the Compressor ON button.

If the error condition still exists, the helium compressor package will not restart.

Error Message	Explanation
"High pressure too High!"	The high side pressure is above the high set point. Release helium from the system.
"High pressure too Low!"	The high side helium below the low set point. Add helium to the system.
“Low Pressure Too Low!”	The helium gas pressure is below the threshold. Add helium to the system.

Error Message	Explanation
“Low Pressure Too High!"	The low side helium pressure is above the high set point; release helium from the system, verify line orientation (lines may be reversed High to Low.)
"Delta Pressure Too High!"	The differential pressure is above the high set point; verify static pressure.
"Delta Pressure Too Low!"	The differential pressure is below the low set point verify static pressure; verify line orientation (lines may be reversed High to Low.)
"Static Pressure Too High!"	The static pressure is above the high set point; verify static pressure, release helium
"Static Pressure Too Low!"	The static pressure is below the low set point; verify static pressure, add helium
"Water In Temp Too High"	The inlet water temperature is above the high set point; reduce water temperature
"Water In Temp Too Low"	The inlet water temperature is below the low set point; increase water temperature
“Water Out Temp Too High!”	The outlet water temperature is above the high set point; increase water flow rate and/or reduce temperature.
"Water Out Temp Too Low"	The outlet water temperature is below the low set point; increase water temperature.
“Helium Temp Too High!”	The helium gas temperature is above the high set point; check cooling water flow rate/temperature, check compressor module oil level by checking sight glass when the helium compressor package is running.
“Helium Temp Too Low!"	Helium discharge temperature below the low set point; check cooling water temperatures, increase ambient temperature if system has been off and ambient temperature is below 40 °F
“Oil Temp Too High!”	The oil temperature is above the high set point; check flow rate and inlet temperature of cooling water.
“Oil Temp Too Low!"	The oil temperature is below the low set point; check cooling water temperatures, increase ambient temperature if system has been off and ambient temperature is below 40 °F (4.4 °C)
“Compressor Motor Current Too Low!”	This occurs when the compressor module motor current is below threshold while the motor is requested running. Can be caused by an overheated compressor module.
“You have had 6 errors in less than an hour: MM:SS”	The helium compressor package is in lockout mode due to more than 6 errors within one hour; the compressor cannot be started until 30 minutes has elapsed, and the lockout has been reset using the reset button. MM:SS indicates minutes : seconds remaining until helium compressor package can be restarted
“3 Phase Power Improperly Wired!”	The order of the phase power is wrong. Re wire the input to the helium compressor package by switching any 2 of the 3 input power wires.
"Power Supply Error!"	Power supply has exhibited a fault, verify voltage output or contact Cryomech, Inc.

7.7.6 Warning diagnostics on display screen

Warnings do not cause the helium compressor package to stop and do not prevent the helium compressor package from starting. If more than one of the warnings below is present, only the highest priority one will be displayed.

Warning Description	Explanation
"High Pressure Running High!"	High pressure is approaching high set point, if the warning does not go away after the cold head cools down, verify that static pressures are correct, and all lines are connected completely.
"High Pressure Running Low!"	High pressure is approaching the low set point; verify static pressure, check historical data to determine if a leak may have occurred.
“Low Pressure Running Low!”	Low pressure is approaching the low set point; verify static pressure, check historical data to determine if a leak may have occurred.
“Low Pressure Running High!"	Low pressure is approaching the high set point; Flex lines may be reversed check orientation of lines, verify that the high pressure is greater than the low pressure
"Delta Pressure Running High!"	Differential pressure approaching high set point; this condition may occur on initial startup. Under normal operation it will reduce as the cold head begins to cool down. If not check static pressure and helium flex line connections
"Delta Pressure Running Low!"	Differential pressure approaching low error set point; helium flex lines may be reversed
"Static Pressure Running High!"	Static pressure approaching high error set point; verify required static pressure before starting the helium compressor package
"Static Pressure Running Low!"	Static pressure approaching low error set point; verify required static pressure before starting the helium compressor package
"Water In Temp Running High"	Water inlet temperature is approaching high error set point; reduce inlet water temperature or increase flow rate to prevent overheating
"Water In Temp Running Low"	Water inlet temperature is approaching low error set point; increase inlet water temperature or reduce flow rate to prevent overcooling or overheating caused by reduced oil temperature
“Water Out Temp Running High!”	Water outlet temperature is approaching high error set point; reduce inlet water temperature or increase flow rate to prevent overheating
"Water Out Temp Running Low"	Water outlet temperature is approaching low error set point; increase inlet water temperature or reduce flow rate to prevent overcooling or overheating caused by reduced oil temperature
“Helium Temp Running Low!"	Helium discharge temperature running low; check cooling water temperatures, increase ambient temperature if system has been off and ambient temperature is below 45 °F

Warning Description	Explanation
“Helium Temp Running High!”	Helium discharge temperature approaching high error set point; check cooling water temperatures, may also be caused by low oil level check compressor module sight glasses or low oil temperature caused by cool water
“Oil Temp Running High!”	Oil temperature approaching high error set point; verify cooling water flow and temperatures
“Oil Temp Running Low!"	Oil temperature approaching low error set point; verify cooling water flow and temperatures (reduce flow rate or increase temperature)

7.7.7 Error and warning set points

The trip and clear set points for all the errors and warnings are listed in the following table.

Error Message	Lock Out Contributor?	Trip	Clear
"High pressure too High!"	YES	400 psig (27.6bar)	375 psig (25.9 bar)
"High pressure too Low!"	YES	150 psig (10.3bar)	170 psig (11.7 bar)
“Low Pressure Too Low!”	YES	40 psig (2.8 bar)	50 psig (3.4 bar)
“Low Pressure Too High!"	YES	250 psig (17.2 bar)	240 psig (16.5 bar)
"Delta Pressure Too High!"	YES	300 psig (20.7bar)	290 psig (20bar)
"Delta Pressure Too Low!"	YES	50 psig (3.4 bar)	75 psig (5.2 bar)
"Static Pressure Too High!"	YES	300 psig (20.7 bar)	280 psig (19.3 bar)
"Static Pressure Too Low!"	YES	100 psig (6.9 bar)	170 psig (11.7 bar)
"Water In Temp Too High"	YES	110 °F (43 °C)	80 °F (27 °C)
"Water In Temp Too Low"	YES	40 °F (4 °C)	50 °F (10 °C)
“Water Out Temp Too High!”	YES	125 °F (52 °C)	110 °F (43 °C)
"Water Out Temp Too Low"	YES	40 °F (4 °C)	50 °F (10 °C)
“Helium Temp Too High!”	YES	190 °F (88 °C)	120 °F (49 °C)
“Helium Temp Too Low!"	YES	40 °F (4 °C)	50 °F (10 °C)
“Oil Temp Too High!”	YES	125 °F (52 °C)	110 °F (43 °C)
“Oil Temp Too Low!"	YES	40 °F (4 °C)	50 °F (10 °C)
“Compressor Motor Current Too Low!”	YES	< 5A	≥ 5 A

Error Message	Lock Out Contributor?	Trip	Clear
“You have had 6 errors in less than an hour: MM:SS”	NO	>6 errors (lock out contributors) in <1 hour	30 minute delay, manual restart
“3 Phase Power Improperly Wired!”	NO	Phase sequence incorrect	Phase sequence correct
"Power Supply Error!"	NO	Power supply fault	Power supply non fault

Warning Description	Trip / Clear
"High Pressure Running High!"	375 psig (25.9 bar)
"High Pressure Running Low!"	170 psig (11.7 bar)
“Low Pressure Running Low!”	50 psig (3.4 bar)
“Low Pressure Running High!"	240 psig (16.5 bar)
"Delta Pressure Running High!"	290 psig (20.0 bar)
"Delta Pressure Running Low!"	75 psig (5.2 bar)
"Static Pressure Running High!"	280 psig (19.3 bar)
"Static Pressure Running Low!"	170 psig (11.7 bar)
"Water In Temp Running High"	85 °F (29 °C)
"Water In Temp Running Low"	45 °F (7 °C)
“Water Out Temp Running High!”	110 °F (43 °C)
"Water Out Temp Running Low"	45 °F (7 °C)
“Helium Temp Running High!”	170 °F (77 °C)
“Helium Temp Running Low!"	45 °F (7 °C)
“Oil Temp Running High!”	120 °F (49 °C)
“Oil Temp Running Low!"	45 °F (7 °C)

7.7.8 Contact Information

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Section 8

Routine Maintenance

Section 8: Routine Maintenance

8.1 Introduction

This section contains maintenance information. For more detailed information, contact Cryomech, Inc.

8.2 Maintenance schedule

Maintenance	Product Type	Frequency	Comment
Replace cold head seals	GM cold heads only	Every 10,000 hours	See Section 8.3
Replace cold head valve and valve plate	GM cold heads only	Every 20,000 hours	See Section 8.3
Replace adsorber	CP100 series only	Every 15,000 hours	See Section 8.4
Replace adsorber	All CP series (except CP100 series)	Every 20,000 hours	See Section 8.4

8.3 Cold head

CAUTION

The Pulse Tube style cold heads contain no user-serviceable parts. Attempting to disassemble the cold head will void the warranty. Contact Cryomech, Inc. if the cold head needs to be returned for servicing.

CAUTION

Service to Gifford-McMahon style cold heads must be performed using Cryomech, Inc. parts and the appropriate work instructions. Contact Cryomech, Inc, before servicing the cold head.

8.4 Replace the adsorber

Required tools CP100

Quantity	Description	Comment
1	1 in open end wrench	For Aeroquip coupling
1	1-1/8 in open end wrench	For Aeroquip coupling
1	1-3/16 in open end wrench	For Aeroquip coupling
1	Slotted screwdriver	For hose clamps
1	Slotted screwdriver	For side panel fasteners

Required tools CP2800

Quantity	Description	Comment
1	1 in open end wrench	For Aeroquip coupling
1	1-1/8 in open end wrench	For Aeroquip coupling
1	1-3/16 in open end wrench	For Aeroquip coupling
1	1-5/8 in open end wrench	For Aeroquip nut
1	Slotted screwdriver	For hose clamps
1	Slotted screwdriver	For side panel fasteners

Required tools CP1100

Quantity	Description	Comment
1	1-3/8 in open end wrench	For Aeroquip coupling
2	1-5/8 in open end wrench	For Aeroquip coupling
1	Slotted screwdriver	For hose clamps
1	Slotted screwdriver	For side panel fasteners

CAUTION

At no time should the Aeroquip couplings be removed from the adsorber. Replacement can be completed without relieving system pressure, since the adsorber is equipped with Aeroquip couplings for sealed removal.

1. Shut down the system.
   a. Press the OFF button on the helium compressor package.
   b. Switch off the circuit breaker mounted on the front-panel to shut down the entire system.
   c. Switch of the dedicated circuit breaker.
   d. Disconnect the input power from the helium compressor package.

WARNING

The input power must be disconnected from the helium compressor package before removing any helium compressor package panels.

2. Disconnect both helium flex lines from the helium compressor package.
3. Remove the screws that hold on the right-side panel and pull the panel away from the helium compressor package.
4. Using three wrenches disconnect the Aeroquip coupling between the adsorber and the oil separator as shown in Figure 8-1.

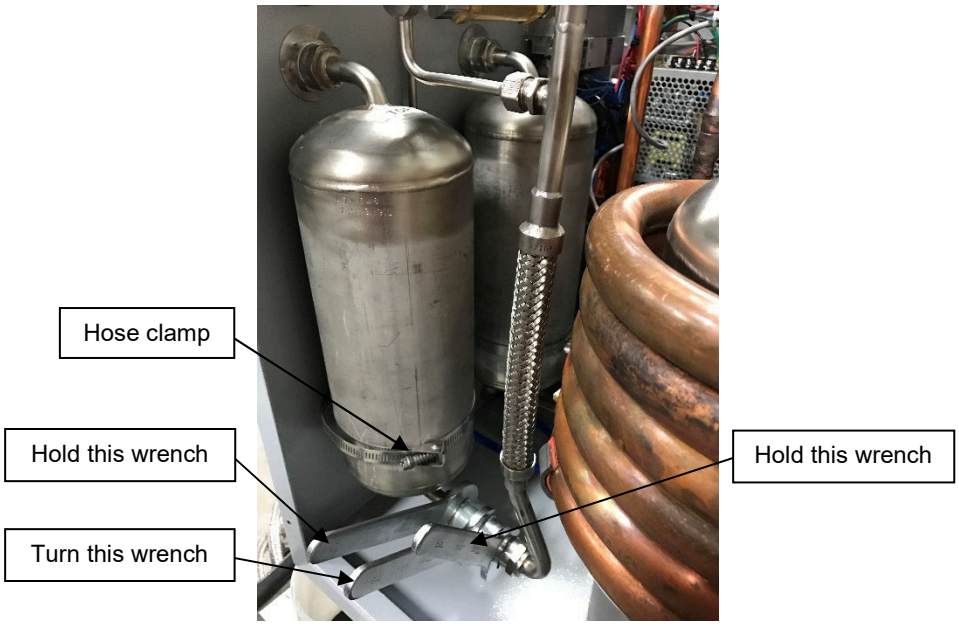


Figure 8-1: Disconnecting the Aeroquip adsorber

5. Remove the nut holding the high pressure Aeroquip coupling to the front panel.
6. Loosen and disconnect the hose clamp that attaches the adsorber to the front panel.
7. Remove the adsorber from the helium compressor package.
8. Check the Aeroquip couplings for oil residue.
   a. If oil is present, contact Cryomech, Inc. for further assistance.
9. Remove the lock washer from the top Aeroquip on CP2800 and CP1100 helium compressor packages.
10. To install the new adsorber, reverse steps 1 through 9.

8.5 Vent excess helium from the helium compressor package

Required tools:

Quantity	Description	Comment
1	3/4 in open end wrench	For Aeroquip coupling
1	Service Aeroquip coupling	For service access port

CAUTION

Venting more than 5 psig (0.34 bar) of helium per minute will lead to improper oil migration within the system. If this condition occurs, factory service will be required.

This procedure should only be used to vent small quantities of helium from an overcharged system.

1. Make sure the service valve is closed. See Figure 8-2.
2. Attach the service Aeroquip coupling to the service access port.

3. Slowly open the service valve.
   - Do not vent more than 5 psig (0.34 bar) of helium per minute.
4. Close the service valve.
5. Remove the service Aeroquip from the service access port.

8.6 Recharge helium to the helium compressor package

Required tools and equipment:

Quantity	Description	Comment
1	3/4 in open end wrench	For Aeroquip coupling
1	Service Aeroquip coupling	For service access port
1	Vacuum/charging system	For adding helium

CAUTION

When adding helium to the helium compressor package, the helium must be 99.999% pure. Contamination of the helium by other gases is a common cause of premature failure and is not covered by the warranty.

1. Turn the helium compressor package off.
2. Allow the entire system (both the helium compressor package and the cold head) to come to room temperature.
3. Use helium with a minimum purity of 99.999%.
4. The helium source and regulator must be capable of pressurizing to the desired pressure.
5. Make sure the service valve is closed.
6. Attach the service Aeroquip coupling to the service access port. See Figure 8-2.

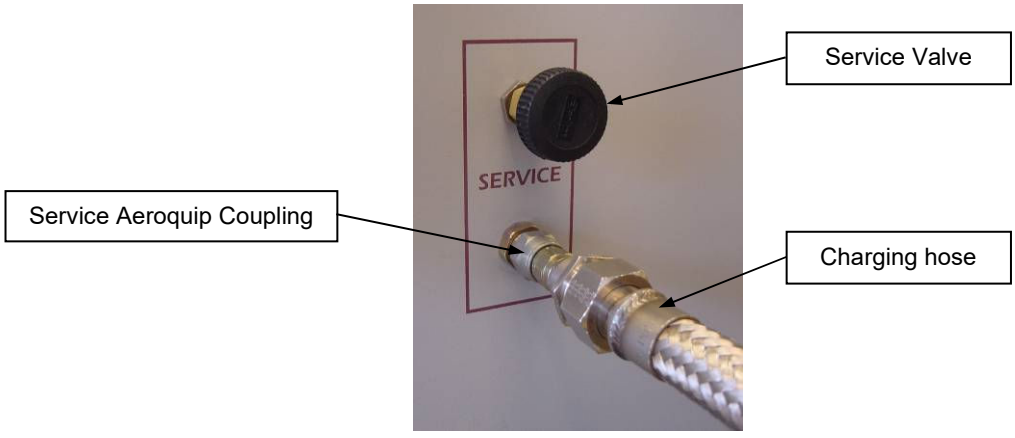


Figure 8-2: Service valve and access port

7. Attach a charging line from the service Aeroquip to a typical vacuum/charging system as shown in Figure 8-3 and Figure 8-4 below.



Figure 8-3: Vacuum/charging system



Figure 8-4: Close up of vacuum/charging manifold and pump

8. Evacuate to 50 microns.
9. Isolate the vacuum pump and add 50 psig (3.4 bar) of helium.
10. Vent the helium.
11. Repeat steps 8, 9 and 10 until the vacuum level reaches 25 microns or less.
12. Pressurize the charging line to the service access with the system helium pressure specified in Section 5.
13. Slowly open the service valve to add helium to the system.
    - The final helium charge in the system helium pressure specified in Section 5.

CAUTION

No more than 5 psig (0.34 bar) of gas should be added per minute to prevent improper oil migration within the system. If such contamination occurs, factory service will be required.

14. Close the service valve.
15. Remove the service Aeroquip from the service access.

8.7 Cleaning

8.7.1 Helium compressor package and cold head

The helium compressor package and cold head require no cleaning other than wiping the outside of each if it becomes dusty or dirty.

CAUTION

Excluding the designated water cooling circuit, never allow water to contact any of the components of the system. Exposing the system to water will void the warranty.

8.7.2 Aeroquip couplings

WARNING

Never remove an Aeroquip coupling from the helium flex lines, cold head, adsorber, or helium compressor package without first relieving the helium charge. Pressure in any of the components can cause the coupling to become a projectile that may cause injury.

If operated in a clean environment, the only parts of the system that are likely to require cleaning are the Aeroquip couplings.

If an Aeroquip coupling needs cleaning:

- Blow off the coupling with clean, dry compressed air.
- Solvents should never be used.
    - If any grease or oil gets on the Aeroquip coupling, contact Cryomech, Inc.